

Mathematica notebook **RLamLGreen**, supplementary material for:

G. Robichon, C. Monthus. W. Krauth “Markov chains at the onset of non-reversibility “,

<https://arxiv.org/abs/2609.05183>

We illustrate basic properties of the transition matrix and the Green's matrix for a 4 x4 example

Here we define the example transition matrix P and compute its eigenvalues. Two eigenvalues are real and a pair of eigenvalues are complex-valued.

```
In[ ]:= P = {{0.05, 0.75, 0. , 0.2 },  
            {0.25, 0.05, 0.7 , 0. },  
            {0, 0.2, 0.05, 0.75},  
            {0.7 , 0. , 0.25, 0.05}}
```

```
Out[ ]:= {{0.05, 0.75, 0. , 0.2}, {0.25, 0.05, 0.7, 0.}, {0, 0.2, 0.05, 0.75}, {0.7, 0. , 0.25, 0.05}}
```

```
In[ ]:= P // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.05 & 0.75 & 0. & 0.2 \\ 0.25 & 0.05 & 0.7 & 0. \\ 0 & 0.2 & 0.05 & 0.75 \\ 0.7 & 0. & 0.25 & 0.05 \end{pmatrix}$$

```
In[ ]:= {λ1, λ2, λ3, λ4} = Eigenvalues[P]
```

```
Out[ ]:= {1. + 0. i, -0.9 + 0. i, 0.05 + 0.497494 i, 0.05 - 0.497494 i}
```

This is the diagonal matrix of eigenvalues.

```
In[ ]:= Λ = {{λ1, 0, 0, 0}, {0, λ2, 0, 0}, {0, 0, λ3, 0}, {0, 0, 0, λ4}}
```

```
Out[ ]:= {{1. + 0. i, 0, 0, 0}, {0, -0.9 + 0. i, 0, 0}, {0, 0, 0.05 + 0.497494 i, 0}, {0, 0, 0, 0.05 - 0.497494 i}}
```

```
In[ ]:= A // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 1. + 0. i & 0 & 0 & 0 \\ 0 & -0.9 + 0. i & 0 & 0 \\ 0 & 0 & 0.05 + 0.497494 i & 0 \\ 0 & 0 & 0 & 0.05 - 0.497494 i \end{pmatrix}$$

```
In[ ]:= Eigenvectors[P] // Chop
```

```
Out[ ]:= {{0.5, 0.5, 0.5, 0.5}, {0.5, -0.5, 0.5, -0.5},
          {0.524404, 0. + 0.474342 i, -0.524404, 0. - 0.474342 i},
          {0.524404, 0. - 0.474342 i, -0.524404, 0. + 0.474342 i}}
```

R is the matrix whose columns are the right eigenvectors

```
In[ ]:= R = Transpose[{{0.5, 0.5, 0.5, 0.5}, {0.5, -0.5, 0.5, -0.5},
                      {0.5244044240850759, 0.474341649025256 i, -0.524404424085075, -0.474341649025256 i},
                      {0.5244044240850759, -0.474341649025256 i, -0.524404424085075, +0.474341649025256 i}}]
```

```
Out[ ]:= {{0.5, 0.5, 0.524404, 0.524404}, {0.5, -0.5, 0. + 0.474342 i, 0. - 0.474342 i},
          {0.5, 0.5, -0.524404, -0.524404}, {0.5, -0.5, 0. - 0.474342 i, 0. + 0.474342 i}}
```

```
In[ ]:= R // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.5 & 0.5 & 0.524404 & 0.524404 \\ 0.5 & -0.5 & 0. + 0.474342 i & 0. - 0.474342 i \\ 0.5 & 0.5 & -0.524404 & -0.524404 \\ 0.5 & -0.5 & 0. - 0.474342 i & 0. + 0.474342 i \end{pmatrix}$$

L is the inverse of R. Its rows contain the complex conjugates of the left eigenvectors

```
In[ ]:= L = Inverse[R]
```

```
Out[ ]:= {{0.5 + 0. i, 0.5 + 0. i, 0.5 + 0. i, 0.5 + 0. i}, {0.5 + 0. i, -0.5 + 0. i, 0.5 + 0. i, -0.5 + 0. i},
          {0.476731 + 0. i, 0. - 0.527046 i, -0.476731 + 0. i, 0. + 0.527046 i},
          {0.476731 + 0. i, 0. + 0.527046 i, -0.476731 + 0. i, 0. - 0.527046 i}}
```

```
In[ ]:= L // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.5 + 0. i & 0.5 + 0. i & 0.5 + 0. i & 0.5 + 0. i \\ 0.5 + 0. i & -0.5 + 0. i & 0.5 + 0. i & -0.5 + 0. i \\ 0.476731 + 0. i & 0. - 0.527046 i & -0.476731 + 0. i & 0. + 0.527046 i \\ 0.476731 + 0. i & 0. + 0.527046 i & -0.476731 + 0. i & 0. - 0.527046 i \end{pmatrix}$$

Here we check that $P = R \Lambda L$, i.e., that P is diagonalized by R and L

```
In[ ]:= Chop[R.Λ.L - P, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= R.L // MatrixForm // Chop
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 1. & 0 & 0 & 0 \\ 0 & 1. & 0 & 0 \\ 0 & 0 & 1. & 0 \\ 0 & 0 & 0 & 1. \end{pmatrix}$$

Here we check some basic linear algebra, namely that $R.L = 1$ implies $L.R = 1$

```
In[ ]:= L.R // MatrixForm // Chop
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 1. & 0 & 0 & 0 \\ 0 & 1. & 0 & 0 \\ 0 & 0 & 1. & 0 \\ 0 & 0 & 0 & 1. \end{pmatrix}$$

Here we break up the matrix R into single-column matrices $R1 + R2 + R3 + R4$ containing each a right eigenvectors

```
In[ ]:= R1 = {{0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}}
```

```
Out[ ]:= {{0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}}
```

```
In[ ]:= {{0.5`, 0, 0, 0}, {0.5`, 0, 0, 0}, {0.5`, 0, 0, 0}, {0.5`, 0, 0, 0}}
```

```
Out[ ]:= {{0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}, {0.5, 0, 0, 0}}
```

```
In[ ]:= R1 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.5 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= R2 = {{0, 0.5, 0, 0}, {0, -0.5, 0, 0}, {0, 0.5, 0, 0}, {0, -0.5, 0, 0}}
```

```
Out[ ]:= {{0, 0.5, 0, 0}, {0, -0.5, 0, 0}, {0, 0.5, 0, 0}, {0, -0.5, 0, 0}}
```

```
In[ ]:= R2 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0.5 & 0 & 0 \\ 0 & -0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & -0.5 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= R3 = {{0, 0, 0.524404, 0}, {0, 0, 0. + 0.47434 i, 0},
               {0, 0, -0.524404, 0}, {0, 0, 0. - 0.47434 i, 0}}
```

```
Out[ ]:= {{0, 0, 0.524404, 0}, {0, 0, 0. + 0.47434 i, 0}, {0, 0, -0.524404, 0}, {0, 0, 0. - 0.47434 i, 0}}
```

```
In[ ]:= R3 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0.524404 & 0 \\ 0 & 0 & 0. + 0.47434 i & 0 \\ 0 & 0 & -0.524404 & 0 \\ 0 & 0 & 0. - 0.47434 i & 0 \end{pmatrix}$$

```
In[ ]:= R4 = {{0, 0, 0, 0.524404}, {0, 0, 0, 0. - 0.47434 i},
               {0, 0, 0, -0.524404}, {0, 0, 0, 0. + 0.47434 i}}
```

```
Out[ ]:= {{0, 0, 0, 0.524404}, {0, 0, 0, 0. - 0.47434 i}, {0, 0, 0, -0.524404}, {0, 0, 0, 0. + 0.47434 i}}
```

```
In[ ]:= R4 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0.524404 \\ 0 & 0 & 0 & 0. - 0.47434 i \\ 0 & 0 & 0 & -0.524404 \\ 0 & 0 & 0 & 0. + 0.47434 i \end{pmatrix}$$

Here we check that the four matrices $R1+R2+R3+R4$ sum up to R and that $P R_m = \lambda_m R_m$.

```
In[ ]:= Chop[R1 + R2 + R3 + R4 - R, 0.0001] // MatrixForm // Chop
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[P . R1 - \lambda1 R1, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[P . R2 - λ2 R2, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[P . R3 - λ3 R3, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[P . R4 - λ4 R4, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Here we break up the matrix L into single-row matrices L1 + L2 + L3 + L4 containing each a single row of L, in other words the conjugate of a left eigenvector. See the main text for a discussion that the rows of L are not the left eigenvectors, but their conjugates.

```
In[ ]:= L1 = {{0.5, 0.5, 0.5, 0.5}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}
```

```
Out[ ]:= {{0.5, 0.5, 0.5, 0.5}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}
```

```
In[ ]:= L1 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Here we check that L1 is the matrix corresponding to the left eigenvector

```
In[ ]:= Chop[L1 . P - λ1 L1, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= L2 = {{0, 0, 0, 0}, {0.5, -0.5, 0.5, -0.5}, {0, 0, 0, 0}, {0, 0, 0, 0}}
```

```
Out[ ]:= {{0, 0, 0, 0}, {0.5, -0.5, 0.5, -0.5}, {0, 0, 0, 0}, {0, 0, 0, 0}}
```

```
In[ ]:= L2 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.5 & -0.5 & 0.5 & -0.5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[L2.P - λ2 L2, 0.001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= L3 = {{0, 0, 0, 0}, {0, 0, 0, 0},
```

```
          {0.476731, -0.527046276694 i, -0.47673129, +0.5270462766947 i}, {0, 0, 0, 0}}
```

```
Out[ ]:= {{0, 0, 0, 0}, {0, 0, 0, 0}, {0.476731, 0. - 0.527046 i, -0.476731, 0. + 0.527046 i}, {0, 0, 0, 0}}
```

```
In[ ]:= L3 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.476731 & 0. - 0.527046 i & -0.476731 & 0. + 0.527046 i \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Note that L3 is a complex matrix, and λ3 a complex eigenvalue. L3 contains the complex conjugate of the third eigenvector.

```
In[ ]:= Chop[L3.P - λ3 L3, 0.001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= L4 = {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0},
```

```
          {0.476731, +0.527046276694 i, -0.47673129, -0.5270462766947 i}}
```

```
Out[ ]:= {{0, 0, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}, {0.476731, 0. + 0.527046 i, -0.476731, 0. - 0.527046 i}}
```

```
In[ ]:= L4 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.476731 & 0. + 0.527046 i & -0.476731 & 0. - 0.527046 i \end{pmatrix}$$

Same as discussed above for L3 and λ_3

```
In[ ]:= Chop[L4 . P -  $\lambda_4$  L4, 0.001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Here we write the product of $R_k L_k$, in other words $|R_k\rangle\langle L_k|$. We start with $R_1 L_1$, also written as $|1\rangle\langle\pi|$. The trace of $R_k L_k$ equals one.

```
In[ ]:= R1.L1 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \\ 0.25 & 0.25 & 0.25 & 0.25 \end{pmatrix}$$

```
In[ ]:= R2.L2 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.25 & -0.25 & 0.25 & -0.25 \\ -0.25 & 0.25 & -0.25 & 0.25 \\ 0.25 & -0.25 & 0.25 & -0.25 \\ -0.25 & 0.25 & -0.25 & 0.25 \end{pmatrix}$$

```
In[ ]:= R3.L3 // MatrixForm // Chop
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.25 & 0. - 0.276385 i & -0.25 & 0. + 0.276385 i \\ 0. + 0.226133 i & 0.249999 & 0. - 0.226133 i & -0.249999 \\ -0.25 & 0. + 0.276385 i & 0.25 & 0. - 0.276385 i \\ 0. - 0.226133 i & -0.249999 & 0. + 0.226133 i & 0.249999 \end{pmatrix}$$

```
In[ ]:= R4.L4 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.25 + 0. i & 0. + 0.276385 i & -0.25 + 0. i & 0. - 0.276385 i \\ 0. - 0.226133 i & 0.249999 + 0. i & 0. + 0.226133 i & -0.249999 + 0. i \\ -0.25 + 0. i & 0. - 0.276385 i & 0.25 + 0. i & 0. + 0.276385 i \\ 0. + 0.226133 i & -0.249999 + 0. i & 0. - 0.226133 i & 0.249999 + 0. i \end{pmatrix}$$

Here, we perform a few checks of orthonormality.

```
In[ ]:= Chop[(R1 + R2 + R3 + R4). (L1 + L2 + L3 + L4), 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0.999999 & 0 & 0 & 0 \\ 0 & 0.999998 & 0 & 0 \\ 0 & 0 & 1. & 0 \\ 0 & 0 & 0 & 0.999998 \end{pmatrix}$$

```
In[ ]:= L1 . R2
```

```
Out[ ]:= {{0., 0., 0., 0.}, {0., 0., 0., 0.}, {0., 0., 0., 0.}, {0., 0., 0., 0.}}
```

```
In[ ]:= R2 . L1
```

```
Out[ ]:= {{0., 0., 0., 0.}, {0., 0., 0., 0.}, {0., 0., 0., 0.}, {0., 0., 0., 0.}}
```

```
In[ ]:= L2 . R2
```

```
Out[ ]:= {{0., 0., 0., 0.}, {0., 1., 0., 0.}, {0., 0., 0., 0.}, {0., 0., 0., 0.}}
```

```
In[ ]:= L3.R3 // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0. + 0. i & 0. + 0. i & 0. + 0. i & 0. + 0. i \\ 0. + 0. i & 0. + 0. i & 0. + 0. i & 0. + 0. i \\ 0. + 0. i & 0. + 0. i & 0.999998 + 0. i & 0. + 0. i \\ 0. + 0. i & 0. + 0. i & 0. + 0. i & 0. + 0. i \end{pmatrix}$$

```
In[ ]:= L2.R3 // MatrixForm // Chop
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```
In[ ]:= Chop[(L1 + L2 + L3 + L4). (R1 + R2 + R3 + R4), 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 1. & 0 & 0 & 0 \\ 0 & 1. & 0 & 0 \\ 0 & 0 & 0.999998 & 0 \\ 0 & 0 & 0 & 0.999998 \end{pmatrix}$$

Here we check the spectral representation of the transition matrix. The matrix PP that we obtain agrees with P.

```
In[ ]:= PP = λ1 R1.L1 + λ2 R2.L2 + λ3 R3.L3 + λ4 R4.L4
```

```
Out[ ]:= {{0.05 + 0. i, 0.75 + 0. i, 2.04599 × 10-8 + 0. i, 0.2 + 0. i},
          {0.250001 + 0. i, 0.0499999 + 0. i, 0.699999 + 0. i, 8.69113 × 10-8 + 0. i},
          {3.56676 × 10-8 + 0. i, 0.2 + 0. i, 0.05 + 0. i, 0.75 + 0. i},
          {0.699999 + 0. i, 8.69113 × 10-8 + 0. i, 0.250001 + 0. i, 0.0499999 + 0. i}}
```

```
In[ ]:= Chop[MatrixForm[%], 0.0001]
```

```
Out[ ]//MatrixForm=
( 0.05      0.75      0      0.2
  0.250001  0.0499999  0.699999  0
    0      0.2      0.05      0.75
  0.699999  0      0.250001  0.0499999 )
```

Here we construct the Green's matrix in terms of powers of the transition matrix P.

```
In[ ]:= G = Sum[MatrixPower[P, k] - R1.L1, {k, 0, 1000}]
```

```
Out[ ]:= {{0.544622, 0.107551, -0.281465, -0.370709}, {-0.327231, 0.544622, 0.0640732, -0.281465},
          {-0.281465, -0.370709, 0.544622, 0.107551}, {0.0640732, -0.281465, -0.327231, 0.544622}}
```

```
In[ ]:= MatrixForm[%]
```

```
Out[ ]//MatrixForm=
( 0.544622  0.107551  -0.281465  -0.370709
 -0.327231  0.544622  0.0640732  -0.281465
 -0.281465  -0.370709  0.544622  0.107551
 0.0640732  -0.281465  -0.327231  0.544622 )
```

Here we check that the Green's matrix commutes with the transition matrix.

```
In[ ]:= Chop[P.G - G.P, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
( 0  0  0  0
  0  0  0  0
  0  0  0  0
  0  0  0  0 )
```

Here we illustrate the spectral representation of the Green's matrix. Clearly, GG is the same as the previously computed G.

```
In[ ]:= GG = 1 / (1 - λ2) R2.L2 + 1 / (1 - λ3) R3.L3 + 1 / (1 - λ4) R4.L4
Out[ ]:= {{0.544622 + 0. i, 0.107551 + 0. i, -0.281464 + 0. i, -0.370709 + 0. i},
          {-0.32723 + 0. i, 0.544621 + 0. i, 0.0640725 + 0. i, -0.281463 + 0. i},
          {-0.281464 + 0. i, -0.370709 + 0. i, 0.544622 + 0. i, 0.107551 + 0. i},
          {0.0640724 + 0. i, -0.281463 + 0. i, -0.32723 + 0. i, 0.544621 + 0. i}}
```

```
In[ ]:= MatrixForm[%] // Chop
Out[ ]//MatrixForm=

$$\begin{pmatrix} 0.544622 & 0.107551 & -0.281464 & -0.370709 \\ -0.32723 & 0.544621 & 0.0640725 & -0.281463 \\ -0.281464 & -0.370709 & 0.544622 & 0.107551 \\ 0.0640724 & -0.281463 & -0.32723 & 0.544621 \end{pmatrix}$$

```

Here we compute the trace of G.

```
In[ ]:= Tr[G]
Out[ ]:= 2.17849
```

...and we check the trace through the Kemeny time.

```
In[ ]:= 1 / (1 - λ2) + 1 / (1 - λ3) + 1 / (1 - λ4) // Chop
Out[ ]:= 2.17849
```

Here we check the orthogonality of the Green's matrix with $\langle \pi | \dots$

```
In[ ]:= Chop[L1.G, 0.0001] // MatrixForm
Out[ ]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

```

and with $|1\rangle$ (the multiplication with L1 and with R1 is a bit overkill, as only one row of L1 and only one column of R1 is non-zero).

```
In[ ]:= Chop[G.R1, 0.0001] // MatrixForm
```

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$